EXPERIMENT NO: -06

AIM OF THE EXPERIMENT: - TO DETERMINE THE OPEN CIRCUIT CHARACTERISTICS (O.C.C) OF A SEPARATELY EXCITED D.C GENERATOR.

THEORY: -

D.C Generator is a device, which converts mechanical energy to electrical energy (D.C in the external circuit due to the rectifying action of the split rings or Commutator & brushes.)

There are 2 types of D.C Generator based on type of excitation; -

1. Self excited (whose field magnets are energized by the power produced by the generator by themselves).
2. Separately Excited (generators whose field magnets are energized from an independent external D.C source).

CHARACTERISTICS

Characteristics of any device represent its functioning and operation. These can be represented by curves or graphs, which are plotted with a specific relationship between various variables.

Among those open circuit characteristics (O.C.C) is one, which is also called magnetic, or no load saturation characteristics. As shown in figure 1.

O.C.C; -

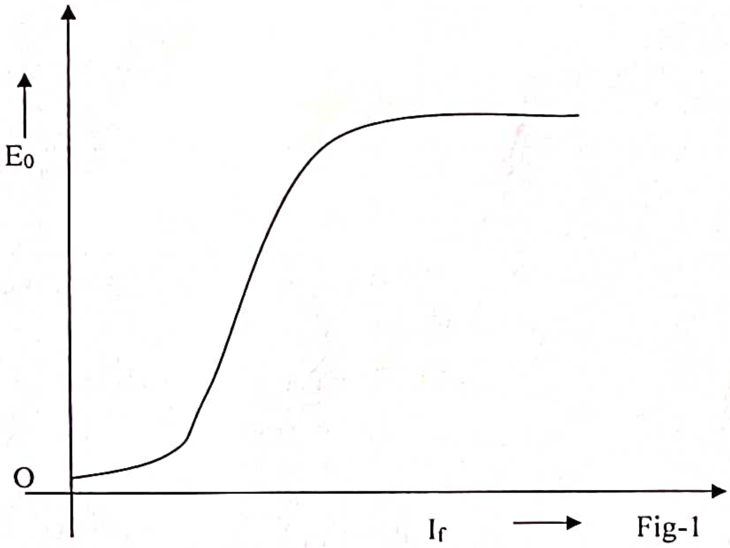


Fig-1

i. It is a curve that represents the relationship between generated emf developed in armature on no load E_o and field flux (current)/ exciting current I_f at a given speed.

$$E_g = \frac{\phi ZN}{60} \times \frac{P}{A} \quad \text{Volts}$$

ii. OCC represents the property for constant speed $E_g = k\Phi = kI_f\, [\Phi \alpha I_f]$ of magnetization for a material of electromagnet. Practically its shape is same for all generators.

iii. The OCC curve does not start from zero point but at some small values slightly higher then zero due to residual magnetism

iv. At one point the field core saturates (even if the field current is increased field flux does not increase) & the curve falls away from the straight line due to varying permeability of iron.

The Field current 'I_f' is varied Rheostatically & its value read on the Ammeter 'A' as shown in figure.1. The machine is driven at constant speed by the prime mover & the generated emf on no-load (The generator terminals open) is measured by the voltmeter connected across the armature terminals.

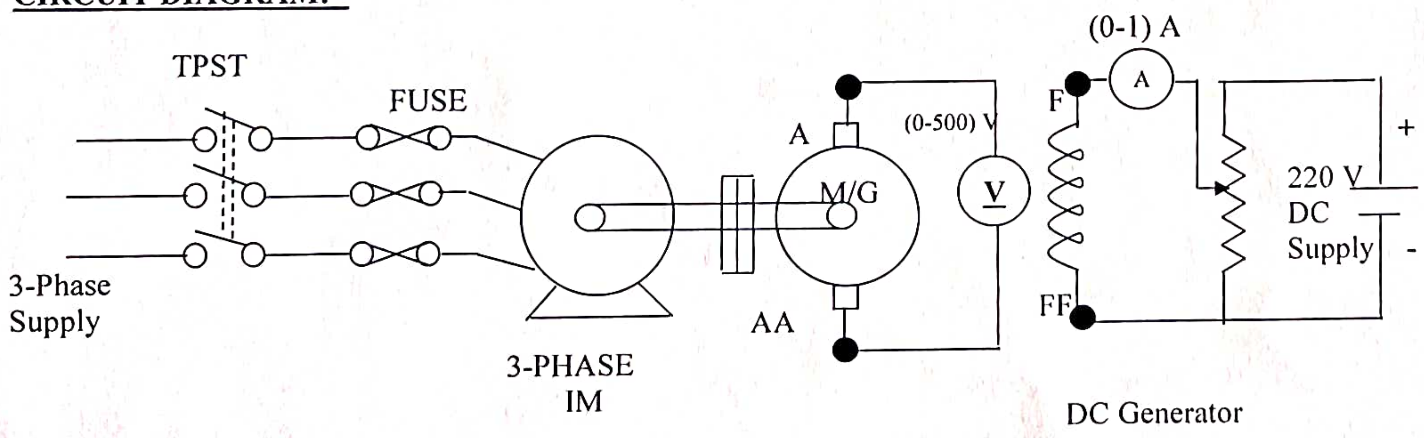
I_f is increased in suitable steps (starting from zero) & the corresponding values of E_o are measured by voltmeter 'V'. On plotting the relation between I_f & E_o, a curve is obtained, known as OCC.

On running the generator at specified speed, due to residual magnetism in the poles, some emf is generated when $I_f = 0$. Hence the curve starts a little way up. The slight curvature at the lower end is due to magnetic inertia. The first part of the curvature is practically straight line, due to the fact that at low flux densities, reluctance of iron path is being negligible (due to high permeability) & the total reluctance is given by the air gap reluctance, which is constant. Hence the flux & consequently, the generated emf is directly proportional to the exciting current. After a point saturation of poles start i.e. reluctance of iron path is prominent & the curve bends down. However the initial slope of the curve is determined by air gap width. The O.C.C for a higher speed would lie above this curve & below it for a lower speed.

APPRATUS REQUIRED: -

SL. NO	APPARATUS	SPECIFICATION	QUANTITY
1	D.C MACHINE	2 KW, 220 V, 9 A	01
2	VOLTMETER	(0-500) V, PMMC TYPE	01
3	AMMETER	(0-1) A, PMMC TYPE	01
4	RHEOSTAT	(0-100)Ω, 5A	01
5	CONNECTING WIRES	3/22 SWG, INSULATED CU WIRES	AS REQUIRED

CIRCUIT DIAGRAM: -



PROCEDURE: -

1. MAKE THE CONNECTION AS PER THE CIRCUIT DIAGRAM SHOWN IN FIG.1
2. SWITCH ON THE A.C SUPPLY AND RUN THE INDUCTION MOTOR.
3. SWITCH ON THE D.C SUPPLY AND GRADUALLY INCREASE THE FIELD VOLTAGE IN SUITABLE STEPS.
4. TAKE THE CORRESPONDING READINGS OF AMMETER AND VOLTMETER.

TABULATION: -

SL. NO.	FIELD CURRENT (I_F) IN AMP.	OPEN CIRCUIT VOLTAGE (E_0) IN VOLT.
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		
9.		
10.		
11.		
12.		

RESULTS: -

Draw the OCC Curve (i.e. I_f Vs E_0) on a graph paper.

CONCLUSION: -

PRECAUTIONS: -

1. ALL THE CONNECTION SHOULD BE RIGHT & TIGHT.
2. THE VOLTMETER & AMMETER SHOULD BE CAREFULLY CHOSEN SO THAT THERE RANGE ARE MORE THAN THE MAXIMUM VALUES TO BE MEASURED.
3. THE FIELD CURRENT SHOULD NOT BE INCRESED TO CROSS THE RATED VALUE.
4. SUPPLY SHOULD BE SWITCHED-ON AFTER ENSURING CORRECTNESS OF CONNECTIONS.
5. ANY LIVE TERMINAL SHOULD NOT BE TOUCHED WHILE SUPPLY IS ON.

QUESTIONS: -

1. What do you mean by a DC generator?
2. What is the function of Commutator in a DC generator?
3. What is residual magnetism and what is its significance in self excited DC generator/
4. Why the OCC is also called magnetic saturation curve?
5. What are the other characteristics of a DC shunt generator?